Does Judge-Panel Agreement Measure Path Invariance? A Measurement-Validity Study of a Claude Agent in Language-Based Economic Games

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Abstract

Outcome metrics in language-based economic games cannot separate consistent strategic reasoning from lucky payoffs. Path-Invariant Agreement (PIA) proposes to fill that gap with Fleiss' κ among LLM judges scoring an agent's reasoning at "equivalent" decision points. We instrument a Claude Sonnet 4.6 GLEE agent to log its reasoning at every decision and ask whether PIA's number measures what it claims. On 105 stratified clusters (805 judged steps, three judge personas) overall κ is 0.007 (game-level 95% CI $[-0.043, 0.079]$). It is below the κ of randomly shuffled clusters in 13 of 14 family–dimension cells, and it is driven by judge calibration offsets and ceiling effects: dropping one lenient judge raises κ from 0.04 to 0.30, and offset-robust agreement is moderate (ICC 0.28–0.70). Observational clusters are also too coarse to define equivalent states: within-cluster bargaining action spread (0.103) is $3.5\times$ the spread measured by re-querying the agent at identical states (0.030; 360 calls). A 20-step check by the author as a human rater agrees with the judge mean on valuation reasoning (Spearman 0.68) but only weakly on horizon/strategy (0.29), and finds the judges more lenient by about 0.9 points. We conclude that cluster-level judge κ is not a valid path-invariance measure here and that same-state replay is the cheaper, sounder consistency probe. We report these negative results together with a corrected account of our earlier submission.

1 Agent, Motivation, and Scope of This Revision

Evaluation of agents in bargaining, negotiation, and persuasion is almost always outcome-based: payoff, win rate, rating. Two agents can reach the same payoff through principled reasoning or through inconsistent reasoning that happened to pay. We built a Claude Sonnet 4.6 agent (GLEE Agent ID `116dce702116`, "PIA-Agent") whose reasoning is logged at every decision, so that a second question can be asked: *is its reasoning consistent across decision points that should be treated alike?*

Path-Invariant Agreement (PIA) answers this by having several LLM judges score reasoning at "equivalent" decision points and reporting Fleiss' κ [Fleiss, 1971]. Our first submission reported $\kappa = 0.0458$ and read it as low path invariance. Reviewers asked whether that number measures path invariance at all. This revision treats that as the research question: we (i) separate three constructs that were conflated, (ii) test the number against within-GLEE baselines, uncertainty estimates, and a small rater check, and (iii) report where it fails. The contribution is a diagnosis of a measurement procedure on a real agent, with data and code released, not a new metric. A point-by-point response to the reviewers of our first submission is appended after the references.

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2 Agent Design and Development

Design. A `strategy(game)` dispatcher routes to per-family handlers sharing one execution core, a choice made under a fixed budget and not because the families are strategically alike. Each call asks Claude for 2–4 sentences of reasoning under `REASONING:` and a JSON action under `ACTION:`; the prompt holds the game text, full machine-readable state, legal-action schema, and round history. Sampling used the API-default temperature, with a two-attempt parse retry and a guaranteed-legal fallback action.

Development. Three failures shaped the data. (1) *Duplicate dispatch*: repeated calls at identical states, first blamed on a polling race, were traced to the agent echoing the schema description instead of a flat action, causing server rejection and re-queueing; fixed with a worked example and a parser unwrap. (2) *Credit exhaustion*: an unattended run entered a multi-hour retry loop; fixed with a circuit breaker and real usage logging. (3) *Prompt caching* gave a 0% hit rate and a 4.7% cost increase and was reverted. Failure (2) makes 69.7% of the first quarter of the log fallback actions (Section 4).

3 Measurement Protocol

Three constructs. For a decision input x (state, legal actions, full history): *same-state repeatability* (SSR) is the spread of actions under repeated sampling from the policy at one fixed x; *path invariance* (PI) requires that policies at inputs x, x' with the same signature σ agree even when their histories differ in features declared irrelevant; *strategic adaptation* is a legitimate difference when the differing history features should update beliefs about the opponent. Observational clustering treats $\sigma(x) = \sigma(x')$ as "equivalent" and assumes the remaining differences are irrelevant. This assumption is untested, and the agent *does* condition on history, so some within-cluster variation is adaptation by design.

Observational protocol. Steps are grouped within a family by a coarse signature: bargaining by round position, pot, and offer/pot ratio; negotiation by round position, role, and price/value ratio; persuasion by round position and bucketed p, v, u. We drew 35 clusters per family (quantile bins over cluster size, at most 10 judged steps per cluster; seed 20261002 and a frozen manifest), giving 805 judged steps. Fallback steps are never judged. Three judge personas (GameTheoreticRigor, LiteralGroundedness, OpponentResponsiveness) score each step on four 1–5 dimensions: valuation reasoning, horizon/strategy, concession handling (not applicable without a prior rejection, hence absent in persuasion), and outcome consistency. The personas share an identical rubric and output schema and differ only in their opening framing. Judge and agent are the same model (Claude Sonnet 4.6) at default temperature.

Statistic. Per cluster, items are judged steps and raters are the three personas; κ is Fleiss' with five categories. The *primary aggregate* is the unweighted mean over clusters; member-weighted and pooled values are sensitivity checks. If all ratings in a cluster are identical, the implementation sets $\kappa = 1$ (denominator ≈ 0). This fires in 12 cluster–dimension entries, all uniform scores of 5, and the primary aggregate excludes them. Clusters sharing a game are not independent (94% of bargaining and 100% of negotiation clusters do), so uncertainty comes from a 10,000-sample bootstrap over *connected components* of the cluster–game graph: 7, 5 and 29 independent units for bargaining, negotiation and persuasion.

4 Results

The headline number is indistinguishable from zero. Table 1 gives the re-computed values. Our first submission's 0.0458 came from a run whose raw judge scores were not persisted; the number here comes from a frozen, reproducible re-run on the same logged trajectories and is the one we stand behind. Negotiation and bargaining intervals include zero. Persuasion is the only family with an interval above zero, yet it sits at the shuffled floor.

Clustering adds nothing to the number. Real κ is below the shuffled mean in 13 of 14 family–dimension cells and above the shuffled 95% interval in none, so arbitrary groups give about the same κ and it cannot be evidence about equivalent states.

	κ all	κ excl. degen.	Game-level 95% CI	Shuffled mean	Shuffled 95%
Bargaining	-0.013	-0.023	$[-0.118, 0.105]$	0.032	$[-0.003, 0.072]$
Negotiation	-0.029	-0.067	$[-0.149, 0.054]$	0.018	$[-0.020, 0.060]$
Persuasion	0.170	0.112	$[0.070, 0.154]$	0.179	$[0.156, 0.204]$
Overall	0.043	0.007	$[-0.043, 0.079]$	–	–

Table 1: Unweighted mean per-cluster Fleiss' κ (35 clusters/family). "Shuffled" reassigns judged steps to random clusters of the same sizes within a family (1,000 permutations; all-clusters version, same forced-κ rule).

	Obs. within-cluster	Replay, default T		Replay, T=0	States
		all states	excl. capped state		
Bargaining	0.103	0.085 [0.023, 0.197]	0.030 [0.021, 0.039]	0.024 / 0.030	5 / 4
Negotiation	0.060	0.090 [0.013, 0.191]	–	0.028 [0, 0.067]	8
Persuasion	0.349	n/a	n/a	n/a	5

Table 2: Mean action spread (SD of offer share, or $1-$modal share). Replay intervals are bootstrap over states. Replay prompts omit the game's rules paragraph (never logged) and reconstruct the action schema.

What κ measures: judge calibration. LiteralGroundedness is systematically lenient, scoring 0.6–1.4 points above the other two judges on horizon and concession handling and up to 0.8 above on bargaining valuation. Several judge pairs have high rank agreement but near-zero κ (negotiation horizon: Spearman 0.71, Cohen's $\kappa = -0.14$). Three checks follow. (i) Per-judge-centered ICC(3,1) is 0.28–0.70 and ordinal Krippendorff's α is 0.13–0.63 (-0.04 for saturated persuasion outcome consistency). (ii) Removing LiteralGroundedness raises overall κ from 0.04 to 0.30 (0.007 to 0.25 excluding degenerate entries), yet two-rater κ is still below its shuffled mean in all 14 cells. (iii) Outcome consistency is saturated: 59% of bargaining and 90% of persuasion scores are 5. Judges do track state: cluster membership explains more judge-score variance than shuffled groups in 6 of 11 cells (negotiation horizon $\eta^2 = 0.52$ vs. 0.15), but κ pooled within clusters does not isolate it. A negative κ means judges disagree beyond chance given their marginals, not that the agent's reasoning is worse than chance. Fleiss' κ is also depressed when items are homogeneous or prevalence is skewed [Feinstein and Cicchetti, 1990]: negotiation clusters are nearly identical in round and phase (modal share 0.99) and give the lowest κ, while persuasion clusters mix rounds (modal round share 0.43) and give the highest. A path-invariance metric should not rise as the compared states become less alike.

Within-GLEE baselines. *Action-only* (within-cluster spread of normalized offers, or $1-$modal share for categorical actions) is no lower than shuffled clusters in bargaining (0.103 vs. 0.122, shuffled 95% $[0.092, 0.157]$) and negotiation (0.060 vs. 0.066, $[0.048, 0.088]$) and lower only in persuasion (0.349 vs. 0.467). Judge-score spread tracks action spread only in persuasion (Spearman 0.51, $p = 0.002$; bargaining 0.04, negotiation -0.01), so in two of three families the judges' score variation carries no information about the agent's behavior variation. We therefore cannot claim that PIA reveals failures that action disagreement misses. We did not run a deterministic heuristic agent or a family-tuned agent.

Same-state replay (SSR). We re-queried the agent on the 18 states in which duplicate dispatch had given identical inputs (16 from iterations before the schema fix), 10 samples each at the API default temperature and at $T = 0$ (360 calls, $2.32). Table 2 gives normalized action spread. In bargaining, the observational within-cluster spread (0.103) is $3.5\times$ the same-state spread (0.030 excluding one state in which 12 of 20 samples reached the 600-token output cap and failed to parse); in negotiation the two are consistent (0.060 vs. 0.090, wide intervals, 8 states). Persuasion is not measurable with our action extractor. Exact action equality fails in 15 of 18 states at *both* settings, but at $T = 0$ the magnitudes are small (0.024–0.028). Together with the baselines above, bargaining clusters therefore mix genuinely different states, which no choice of judge can repair.

Human rater check ($n = 20$). The first author rated 20 randomly chosen judged steps (7/7/6 by family; sheet showed no judge scores) on three dimensions. Rank correlation with the judge mean is 0.68 [0.43, 0.84] for valuation reasoning but 0.29 $[-0.20, 0.68]$ for horizon/strategy, and the judges are more lenient than the rater by 0.87 and 0.85 points. Several low horizon ratings concern games

whose horizon is unknown, a case our rubric does not address. With one rater who knew the study's hypothesis and $n = 20$, this is a sanity check on the judges, not validation, and no human–human agreement was measured.

Fallback rate. Of 5,870 steps in the sampled clusters, 883 (15.0%) were fallback actions (bargaining 10.8%, negotiation 4.5%, persuasion 21.5%); none of the 805 judged steps is a fallback. Over the whole 26,319-step log the rate is 20.8%, almost entirely from the credit-exhaustion episode (69.7% in the first quartile of the log, 0.2% in the third, none in the last).

5 Agent Behavior Analysis

Design intent and verification. The agent was designed to compute reservation values and equilibrium benchmarks, model the opponent from history, and act consistently with that reasoning. We verified this against logged reasoning and actions, not against win rate. The evidence is mixed. Judges rate outcome consistency high (though the scale is saturated), but the step from stated reasoning to concrete action is unstable. At a \$1M, 12-round bargaining state ($\delta_1 = 0.9$), 8 of 10 default-temperature replay samples cite the same equilibrium share ($1/(1+\delta) \approx 52.6\%$, assuming Bob's δ equals the agent's) yet propose 52.6% to 65% for themselves, justified as "a strong opening offer"; at $T = 0$, 8 of 10 cite it and offer 52.6% to 60%. The variation is introduced by a heuristic anchoring step, not by the equilibrium computation or by rounding. The archived pre-fix calls at a $\delta = 0.95$ state showed a related pattern but with divergent reasoning: of four calls, one derived $1/(1+\delta) = 51.3\%$ and the others reasoned heuristically (offers of 55% twice and 60% twice). We withdraw our earlier statement that all four derived the same solution.

Faithfulness. PIA scores the consistency of *reported* rationales; these may be post-hoc explanations [Turpin et al., 2023, Lanham et al., 2023] and do not establish consistency of the underlying computation. The outcome-consistency dimension checks only that the action follows from the stated reasoning.

6 Limitations and Reproducibility

Equivalence is untested. We did not sweep signature strictness; shuffled clusters (loosest) and near-identical negotiation clusters (strictest) bracket it. Persuasion clusters mix rounds and histories, so they are the weakest claim to equivalence. **Judges.** All three share one base model with the agent, so self-preference [Zheng et al., 2023, Panickssery et al., 2024] may contribute to leniency; judges were sampled once, so judge test–retest reliability is unknown. **Scale.** The replay has 5–8 states per family, and per-family intervals are wide. **Prompt reconstruction.** Replay prompts omit the rules paragraph, and parse failures concentrate in one state. **Withdrawn comparison.** Earlier versions compared GLEE κ with a wearable-domain κ of 0.743. That comparison was invalid because κ depends on prevalence and cluster construction, and the underlying figures came from scripted dry-run scores, not from live raters; we withdraw them. **Novelty.** PIA's comparison mechanism comes from the author's prior work [Bade, 2026]; this paper's contribution is domain transfer and its diagnosis.

Reproducibility. Code, the frozen manifest (input hash 85ff4578…), raw per-judge scores, replay samples, ratings and all analysis outputs are at `https://github.com/finaspirant/llm-wearable-agentic-eval-pipeline` (`src/glee/`, `src/rsi_bench/`, `data/rsi_bench/`). Analyses use seed 20261002 and make no API calls. The scoring run made 2,415 judge calls (about \$17.5). Our earlier submission's driver script was not in the repository at submission time; the scripts released now regenerate every number in this paper.

Use of Large Language Models

The submitted agent and the judges are LLM-based (Claude Sonnet 4.6) and are the objects of study. Implementation and analysis code were developed with an AI coding assistant (Claude Code) and this manuscript was drafted with AI assistance, under the author's supervision; the author reviewed the experiments and takes responsibility for all claims. Every number was recomputed from the released analysis files.

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Response to Reviewers: Revise & Resubmit

Does Judge-Panel Agreement Measure Path Invariance? A Measurement-Validity Study of a Claude Agent in Language-Based Economic Games (IAB@NeurIPS 2026 Competition Track, submission #2)

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We thank both reviewers. Reviewer FzAs (R2) in particular gave a precise, correct diagnosis, and we have made it the organizing question of the revision: **does the PIA number measure path invariance in GLEE at all?** We ran the checks the reviewers asked for, and most of them went against our original claims. The revised paper therefore reports a different headline, retracts several statements, and repositions the contribution as a measurement-validity study. We would rather submit an honest negative result than defend a number the evidence does not support.

1. Summary of what changed

	First submission	Revision
Headline	Overall $\kappa = 0.0458$, read as "slight agreement" and low path invariance	Overall $\kappa = 0.043$ with degenerate entries, **0.007 excluding them**; game-level 95% CI [−0.043, 0.079]. We conclude the number is not a valid path-invariance measure here
Uncertainty	None	Cluster-dependence-aware bootstrap over connected components of the cluster–game graph (7 / 5 / 29 independent units for bargaining / negotiation / persuasion)
Baselines	None within GLEE	Shuffled-cluster floor (1,000 permutations), action-only baseline, same-state replay (360 calls), leave-one-judge-out, offset-robust agreement
Judge validation	None	Judge-calibration diagnostics, plus a 20-step single-rater check (a sanity check, not validation)
Constructs	Replay and clustering conflated	Same-state repeatability, path invariance, and strategic adaptation defined separately (Sec. 3)
Cross-domain comparison ($\kappa = 0.743$ vs. 0.0458)	Presented as a headline contrast	**Withdrawn** (Sec. 6)
Rubinstein example	"Four calls derived the same solution"	**Withdrawn**; replaced with new, verified replay evidence (Sec. 5)
Fallback rate	Not reported	Reported (Sec. 4)
Contribution	New process-level metric	Domain transfer and empirical diagnosis of an existing procedure

Corrections we owe the reviewers.

1. *The 0.0458 figure.* Its raw judge scores were not persisted, and the driver script was not in the repository at submission time. We therefore re-ran the scoring from the logged trajectories under a frozen manifest (seed 20261002, 35 clusters per family, 805 judged steps, 2,415 judge calls, about $17.5). All numbers in the revision come from this reproducible run. We do not claim the new number confirms the old one.
2. *The 0.743 wearable-domain figure.* The underlying scores were scripted dry-run scores, not live-rater scores. We withdraw it, and we also withdraw the −0.065 naive-baseline figure that R2 cited.
3. *The Rubinstein example.* The saved archived calls do not support "all four derived the same solution." Only one of four derived 51.3%; the others reasoned heuristically. See M7 below.

Every numeric claim in the paper (104 of them) is checked against the released analysis files by a script, `verify_numbers.py`, which reports 0 mismatches.

2. Reviewer FzAs (R2): major concerns

M1. "The main observational measurement does not cleanly isolate path invariance" (approximate clustering conflates inconsistency with legitimate adaptation). *Agreed, and our new results show the problem is worse than the reviewer assumed.* We now separate the constructs (Sec. 3) and test whether clustering contributes anything. (a) **Shuffled-cluster floor.** Reassigning judged steps to random same-size clusters within a family gives κ of 0.032 (bargaining), 0.018 (negotiation) and 0.179 (persuasion). Real κ is below the shuffled mean in 13 of 14 family–dimension cells and above the shuffled interval in none. So the cluster structure carries no detectable signal in κ. (b) **Action spread.** Within-cluster action spread in bargaining (0.103) and negotiation (0.060) is no lower than for shuffled clusters (0.122 and 0.066), so the clusters do not group actions that are more alike than random. (c) **Replay.** Bargaining action spread at *identical* states is 0.030 (excluding one parse-failure state), 3.5× smaller than the observational 0.103, so bargaining clusters mix genuinely different states. We conclude that the coarse signatures do not define equivalent states and that κ over them cannot separate inconsistency from adaptation. We say so in the abstract.

M2. "The controlled replay tests stochastic repeatability, not path dependence." *Agreed.* The paper now names it **same-state repeatability (SSR)** and defines it separately from path invariance and strategic adaptation (Sec. 3). Nowhere do we describe replay as path invariance. We also expanded it: 18 states × 2 temperatures × 10 samples (360 calls), with bootstrap intervals over states (Table 2). We do not run a true path-invariance experiment (identical state, different irrelevant histories), because that needs a verified irrelevance relation, which we do not have; this is stated as a limitation.

M3. "The cross-domain κ comparison (0.0458 vs. 0.743) is difficult to interpret." *Agreed; withdrawn.* κ depends on prevalence, rubric, cluster construction and judge behavior, all of which differ between domains. We also disclose that the 0.743 came from scripted dry-run scores (Sec. 6). The paper makes no cross-domain claim. We note one direct consequence of R2's point that we measured: κ rises as compared states become *less* alike (persuasion clusters, modal round share 0.43, have the highest κ; near-identical negotiation clusters, modal share 0.99, have the lowest), which is the prevalence/homogeneity effect (Feinstein & Cicchetti, 1990).

M4. "No meaningful within-GLEE baseline." *Done, except two items.* Of the six baselines R2 listed:

- *action-only agreement*: done (Sec. 4);
- *randomly shuffled clusters*: done;
- *repeated calls at identical states*: done (SSR, Table 2);
- *exact or lexical reasoning agreement*: not done; we replaced it with judge-offset-robust agreement (ICC(3,1) 0.28–0.70, ordinal Krippendorff's α 0.13–0.63);
- *deterministic heuristic agent*: **not done**;
- *family-specific prompted agent*: **not done** (see "What we did not do" below).

The baselines change the interpretation: 0.007 is not "surprisingly low," it is what arbitrary groupings and judge offsets produce.

M5. "The reliability of the LLM judges is not established." *Partly addressed, and we state the limits.* The requested four items:

- *Sensitivity to judge choice* — done. One judge (LiteralGroundedness) is systematically lenient, scoring 0.6–1.4 points above the others on horizon and concession handling. Dropping it raises overall κ from 0.043 to 0.305 (0.007 to 0.253 excluding degenerate entries). Judge pairs can have high rank agreement but near-zero κ (negotiation horizon: Spearman 0.71, Cohen's κ −0.14). Much of what κ captures is judge calibration, not agent behavior.
- *Confidence intervals* — done (below, M6).
- *LLM–human agreement* — partly done: the first author rated 20 randomly chosen judged steps (7/7/6 by family; the sheet showed no judge scores). Rank correlation with the judge mean is 0.68 [0.43, 0.84] on valuation reasoning but 0.29 [−0.20, 0.68] on horizon/strategy; the judges are more lenient by about 0.9 points on both. Outcome consistency was constant (5) on all 20.
- *Human–human agreement* — **not done.** This is a single rater who knew the study's hypothesis, with n = 20. We call it a sanity check, not validation, and the paper says so.

Our answer to the reviewer's question (could the negative κ come from evaluator disagreement?) is: yes, substantially. Judge offsets and ceiling effects (59% of bargaining and 90% of persuasion outcome-consistency scores are 5) explain much of it, and a negative κ means judges disagree beyond chance given their marginals, not that the agent's reasoning is worse than chance (see m5).

M6. "Uncertainty on the reported κ values is missing." *Done.* A naive cluster bootstrap would understate uncertainty because clusters share games (94% of bargaining and 100% of negotiation clusters do). We therefore bootstrap over connected components of the cluster–game graph (10,000 samples). Results (Table 1): overall [−0.043, 0.079]; bargaining [−0.118, 0.105]; negotiation [−0.149, 0.054]; persuasion [0.070, 0.154]. Per-dimension values and intervals are in the released analysis files. The bargaining and negotiation intervals include zero, so we now claim only that these data cannot distinguish them from the shuffled floor, which is a weaker statement than the first submission's "statistically indistinguishable from chance" interpretation.

M7. "Faithfulness of the reasoning trace." *Agreed.* Sec. 5 now states that PIA scores the consistency of *reported* rationales, which may be post-hoc (Turpin et al., 2023; Lanham et al., 2023), and that it does not establish consistency of the latent computation. The outcome-consistency dimension checks only that the

action follows from the stated reasoning, and we report that it is saturated, so it is weak evidence even for that.

M8. "Does PIA add information beyond simpler stability metrics?" *Honest answer: we found no evidence that it does, and the paper says so.* Judge-score spread tracks action spread only in persuasion (Spearman 0.51, $p = 0.002$); in bargaining (0.04) and negotiation (−0.01) the judges' score variation carries no information about behavioral variation. We did not find a failure that PIA reveals and action disagreement misses. We therefore recommend same-state action replay as the cheaper and sounder first-line probe. The one thing the judge panel did show is that cluster membership explains more judge-score variance than shuffled groups in 6 of 11 cells (negotiation horizon $\eta^2 = 0.52$ vs. 0.15); that is, judges do respond to state differences, but κ pooled within clusters does not isolate that response. We did not test whether low PIA predicts poor outcomes.

M9. "The adaptation is incremental relative to prior work." *Agreed; repositioned.* The title, abstract and Sec. 1 now present the paper as domain transfer plus empirical diagnosis, "not a new metric," and Sec. 6 states that the comparison mechanism comes from the author's prior work.

3. Reviewer FzAs (R2): minor comments

m1. Formal definition of "equivalent decision points." Sec. 3 now defines the three constructs over a decision input x (state, legal actions, full history) and states what the observational protocol assumes: $\sigma(x) = \sigma(x')$ implies the remaining differences are irrelevant. We state explicitly that this assumption is untested and that the agent conditions on history, so some variation is adaptation by design. The empirical result (M1) is that the assumption fails for the signatures used.

m2. Sampling configuration. The agent and judges used the API-default temperature with no other decoding changes; judges were sampled once (Sec. 3). Replay was run at both the default and $T = 0$.

m3. Rubinstein derivation. New and corrected (Sec. 5). We reran the \$1M, 12-round bargaining state ($\delta_1 = 0.9$): in 8 of 10 default-temperature samples the agent cites the same equilibrium share ($1/(1+\delta) \approx 52.6\%$, assuming Bob's δ equals its own) yet proposes 52.6% to 65% for itself, justified as a "strong opening offer." At $T = 0$, 8 of 10 cite it and offer 52.6% to 60%. So the variation enters at a heuristic anchoring step, not in the equilibrium computation and not in rounding. This supports R2's hypothesis about action translation, with the caveat that it is one state. For the archived $\delta = 0.95$ calls, the saved text shows one call derived 51.3% and the others reasoned heuristically; we withdraw the earlier "all four" claim.

m4. High- vs. low-agreement qualitative examples. *Not added as examples.* The space went to the quantitative explanation of why persuasion is higher: its clusters mix rounds, and its judge spread tracks action spread, while negotiation clusters are nearly identical (modal share 0.99). We are happy to add examples to the camera-ready version if requested.

m5. "Worse than chance" wording. Fixed. The paper now says a negative κ means judges disagree beyond chance given their marginals, not that the agent's reasoning is worse than chance (Sec. 4).

m6. Family-specific prompting ablation. Not done. See "What we did not do."

4. Reviewer FzAs (R2): questions

Question	Answer
What makes two states equivalent?	Sec. 3 (m1). Same coarse signature, with remaining differences assumed irrelevant. We show this assumption does not hold for bargaining.
How much does PIA change with stricter/looser signatures?	Not swept. The shuffled clusters (loosest) and near-identical negotiation clusters (strictest) bracket it, and κ is at or below the shuffled floor in both regimes. A signature-strictness sweep remains future work.
Bootstrap CIs?	M6. Aggregate and per-family are in Table 1; per-dimension are in the released analysis files.
Judge reliability vs. human raters?	M5. Single rater, n = 20, and a sanity check only. Yes, the negative κ values plausibly arise mainly from evaluator disagreement.
Decoding temperature? How much of the divergence vanishes at T ≈ 0?	m2. Exact action equality fails in 15 of 18 states at *both* settings, but at T = 0 the magnitudes are small (spread 0.024–0.028). So the divergence is mostly small-magnitude variation, not large disagreement. (Our re-run finds 15 of 18 states, versus 16 of 18 in the first submission; the re-run uses 10 samples per state and the earlier count came from fewer calls, so we do not read the difference as meaningful.)
Why interpret exact-state replay as path invariance?	We do not. It is SSR (M2).
Within-GLEE baseline?	M4.
Does PIA identify failures not revealed by repeated-action agreement or outcomes?	M8. No evidence that it does.
Sensitivity to each judge, dimension, and large clusters?	Judge: leave-one-judge-out (M5). Dimension: per-dimension results and ICC/α ranges. Large clusters: the unweighted mean is the primary aggregate, with member-weighted and pooled values reported as sensitivity checks. We did not run a leave-largest-cluster-out analysis.

5. Reviewer t66e09 (R1)

C1. Unreported fallback rate (recommendation: include in Sec. 6). *Done* (Sec. 4, “Fallback rate”). Of 5,870 steps in the sampled clusters, 883 (15.0%) were fallbacks (bargaining 10.8%, negotiation 4.5%, persuasion 21.5%). None of the 805 judged steps is a fallback. Over the whole 26,319-step log the rate is 20.8%, almost entirely from the credit-exhaustion episode (69.7% in the first quartile of the log, 13.1% in the second, 0.2% in the third, none in the fourth).

C2. Naive/heuristic baseline for GLEE. *Partly done.* Shuffled-cluster, action-only, and same-state replay baselines are in (M4). A deterministic heuristic *agent* was not run. We believe the three baselines answer R1’s actual question (whether 0.007 reflects the agent, the prompt structure or the clustering): it reflects the judging and clustering procedure, since arbitrary groupings give the same value.

C3. Negative-κ dimensions (human calibration or prompt tweaks). *Addressed, with a limit.* The horizon and concession negative κ values are explained mainly by a lenient judge and the resulting calibration offsets (M5): dropping that judge moves overall κ from 0.043 to 0.305, and offset-robust agreement is moderate (ICC 0.28–0.70). The 20-step rater check found weak agreement with judges on horizon (0.29 [−0.20, 0.68]), and several of the author's low horizon ratings concerned games with an unknown horizon, a case the rubric does not address. A rubric with anchored examples and an explicit unknown-horizon rule is the natural prompt fix; we have not tested it.

R1's strength about the Rubinstein case. R1 listed the identical-derivation Rubinstein example as compelling. We have corrected it (m3) and the replacement evidence is, we think, stronger and verifiable.

6. What we did not do

We prefer to list these plainly rather than leave reviewers to find them.

1. **Signature-strictness sweep** (R2 minor, Q2). Not run.
2. **Deterministic heuristic agent** and **family-specific prompted agent** (R1 C2; R2 baselines and m6). Not run. The paper says so in Sec. 4 and Sec. 6.
3. **Human–human agreement and a multi-rater human study** (R2 M5). We have one rater, n = 20, who knew the hypothesis.
4. **A true path-invariance experiment** with a verified irrelevance relation (R2 M2).
5. **Qualitative high- vs. low-agreement examples** (R2 m4).
6. **Judge test–retest reliability and non-Claude judges.** Judges share a base model with the agent, so self-preference (Zheng et al., 2023; Panickssery et al., 2024) may contribute to the leniency; each judge was sampled once.
7. **Replay scale and fidelity.** 5–8 states per family; persuasion is not measurable with our action extractor; replay prompts reconstruct the action schema and omit the game's rules paragraph; 12 of 13 parse failures fall in one bargaining state.

7. Resulting claims

We believe the revised paper makes four claims that the evidence supports: (1) in this GLEE setting, cluster-level judge-panel κ is at or below an arbitrary-grouping floor and is dominated by judge calibration and ceiling effects, so it is not a valid path-invariance measure; (2) the coarse observational signatures do not define equivalent states in bargaining; (3) same-state replay is a cheaper and sounder consistency probe, and it shows that action variation can enter after a stable equilibrium computation; (4) the agent's behavior, the data, and the code are released so these results can be rechecked. We have removed the claims that did not survive our own checks, and we thank the reviewers for pushing us to find them.